



2 - Propositional Logic + Truth Table

Week 3 – 26 Aug 2025

Tutor: Chin Yi

Cheat sheet

- Commutative laws:

$$p \wedge q \equiv q \wedge p$$

$$p \vee q \equiv q \vee p$$

- Associative laws:

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

- Distributive laws:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

- Identity laws:

$$p \wedge \mathbf{T} \equiv p$$

$$p \vee \mathbf{F} \equiv p$$

- Negation laws:

$$p \vee \neg p \equiv \mathbf{T}$$

$$p \wedge \neg p \equiv \mathbf{F}$$

- Double negation:

$$\neg(\neg p) \equiv p$$

- Idempotent laws:

$$p \wedge p \equiv p$$

$$p \vee p \equiv p$$

- Universal bound laws:

$$p \vee \mathbf{T} \equiv \mathbf{T}$$

$$p \wedge \mathbf{F} \equiv \mathbf{F}$$

- De Morgan's laws:

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

- Absorption laws:

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

- Negations of **T** and **F**:

$$\neg \mathbf{T} \equiv \mathbf{F}$$

$$\neg \mathbf{F} \equiv \mathbf{T}$$

Cheat sheet

Modus Ponens	$p \rightarrow q$ p $\therefore q$	Elimination	$p \vee q$ $\neg q$ $\therefore p$	$p \vee q$ $\neg p$ $\therefore q$
Modus Tollens	$p \rightarrow q$ $\neg q$ $\therefore \neg p$	Transitivity	$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$	
Generalization	p $\therefore p \vee q$	Proof by division into cases	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\therefore r$	q $\therefore p \vee q$
Specialization	$p \wedge q$ $\therefore p$			
Conjunction	p q $\therefore p \wedge q$	Contradiction Rule	$\neg p \rightarrow \mathbf{F}$ $\therefore p$	

Ex. 2.1.22, 24 Determine which of the pairs of statement forms are logically equivalent. Justify your answers using truth tables and include a few words of explanation.

22. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$.

24. $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$.

22. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$.

p	q	r	$q \vee r$	$p \wedge q$	$p \wedge r$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
1	1	1	1	1	1	1	1
1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

same truth values

24. $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$.

p	q	r	$p \vee q$	$p \wedge r$	$(p \vee q) \vee (p \wedge r)$	$(p \vee q) \wedge r$
1	1	1	1	1	1	1
1	1	0	1	0	1	0
1	0	1	1	1	1	1
1	0	0	1	0	1	0
0	1	1	1	0	1	1
0	1	0	1	0	1	0
0	0	1	0	0	0	0
0	0	0	0	0	0	0

←

←

←

different truth values

Ex. 2.1.28. Use De Morgan's laws to write a negation for the statement

“This computer program **has** a logical error in the first ten lines **or** it is **being run** with an incomplete data set.”

SOLUTION . This computer program does not have a logical error in the first ten lines and it is not being run with an incomplete data set.

Note the statement “This computer program does not have a logical error in the first ten lines and it is being run with a complete data set.” is not an acceptable answer. The negation of “it is being run with an incomplete data set” is not “it is being run with a complete data set”. For example, it could be the case that the program is not running at all. □

Ex. 2.1.33. Assume x is a particular real number and use De Morgan's laws to write negations for the statement

$$-10 < x < 2. \quad \text{Equivalent to } -10 < x \text{ AND } x < 2$$

SOLUTION .

$$-10 \geq x \quad \text{or} \quad x \geq 2.$$

(Here we are assuming $x \in \mathbb{R}$.)



Ex. 2.1.42. Use a truth table to establish if the statement form is a tautology or a contradiction.

$$((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q.$$

SOLUTION . The truth table can be done like this:

p	q	r	$\neg p$	$\neg p \wedge q$	$q \wedge r$	$\neg q$	$(\neg p \wedge q) \wedge (q \wedge r)$	$((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$
1	1	1	0	0	1	0	0	0
1	1	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0
1	0	0	0	0	0	1	0	0
0	1	1	1	1	1	0	1	0
0	1	0	1	1	0	0	0	0
0	0	1	1	0	0	1	0	0
0	0	0	1	0	0	1	0	0


 all 0's

Since all the truth values of $((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$ are 0's, $((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$ is a contradiction. □

Ex. 2.1.49. Supply a reason for each step.

$$\begin{aligned}(p \vee \neg q) \wedge (\neg p \vee \neg q) &\equiv (\neg q \vee p) \wedge (\neg q \vee \neg p) && \text{by (a)} \\ &\equiv \neg q \vee (p \wedge \neg p) && \text{by (b)} \\ &\equiv \neg q \vee \mathbf{F} && \text{by (c)} \\ &\equiv \neg q && \text{by (d)}\end{aligned}$$

Therefore, $(p \vee \neg q) \wedge (\neg p \vee \neg q) \equiv \neg q$.

- Commutative laws:

$$\begin{aligned}p \wedge q &\equiv q \wedge p \\ p \vee q &\equiv q \vee p\end{aligned}$$

- Associative laws:

$$\begin{aligned}(p \wedge q) \wedge r &\equiv p \wedge (q \wedge r) \\ (p \vee q) \vee r &\equiv p \vee (q \vee r)\end{aligned}$$

- Distributive laws:

$$\begin{aligned}p \wedge (q \vee r) &\equiv (p \wedge q) \vee (p \wedge r) \\ p \vee (q \wedge r) &\equiv (p \vee q) \wedge (p \vee r)\end{aligned}$$

- Identity laws:

$$\begin{aligned}p \wedge \mathbf{T} &\equiv p \\ p \vee \mathbf{F} &\equiv p\end{aligned}$$

- Negation laws:

$$\begin{aligned}p \vee \neg p &\equiv \mathbf{T} \\ p \wedge \neg p &\equiv \mathbf{F}\end{aligned}$$

- Double negation:

$$\neg(\neg p) \equiv p$$

- Idempotent laws:

$$\begin{aligned}p \wedge p &\equiv p \\ p \vee p &\equiv p\end{aligned}$$

Ex. 2.1.52. Use the logical laws to verify the logical equivalence

$$\neg(p \vee \neg q) \vee (\neg p \wedge \neg q) \equiv \neg p.$$

Supply a reason for each step.

- **Commutative laws:**

$$p \wedge q \equiv q \wedge p$$

$$p \vee q \equiv q \vee p$$

- **Associative laws:**

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

- **Distributive laws:**

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

- **Identity laws:**

$$p \wedge \mathbf{T} \equiv p$$

$$p \vee \mathbf{F} \equiv p$$

- **Negation laws:**

$$p \vee \neg p \equiv \mathbf{T}$$

$$p \wedge \neg p \equiv \mathbf{F}$$

- **Double negation:**

$$\neg(\neg p) \equiv p$$

- **Idempotent laws:**

$$p \wedge p \equiv p$$

$$p \vee p \equiv p$$

- **Universal bound laws:**

$$p \vee \mathbf{T} \equiv \mathbf{T}$$

$$p \wedge \mathbf{F} \equiv \mathbf{F}$$

- **De Morgan's laws:**

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

- **Absorption laws:**

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

- **Negations of $\mathbf{T}$ and $\mathbf{F}$:**

$$\neg \mathbf{T} \equiv \mathbf{F}$$

$$\neg \mathbf{F} \equiv \mathbf{T}$$

Ex. 2.2.12 (Modified). Apply Ex. 2.2.13a (below) to establish the logical equivalence

$$p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r).$$

Next, use it rewrite the following statement. (Assume that x represents a fixed real number.)

If $x > 2$ or $x < -2$, then $x^2 > 4$.

Ex. 2.2.13a. $p \rightarrow q \equiv \neg p \vee q.$

Ex. 2.2.13a. $p \rightarrow q \equiv \neg p \vee q$.

$$p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r).$$

- Commutative laws:

$$p \wedge q \equiv q \wedge p$$

$$p \vee q \equiv q \vee p$$

- Associative laws:

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

- Distributive laws:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

- Identity laws:

$$p \wedge \mathbf{T} \equiv p$$

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- Negation laws:

$$p \vee \neg p \equiv \mathbf{T}$$

$$p \wedge \neg p \equiv \mathbf{F}$$

- Double negation:

$$\neg(\neg p) \equiv p$$

- Idempotent laws:

$$p \wedge p \equiv p$$

$$p \vee p \equiv p$$

- Universal bound laws:

$$p \vee \mathbf{T} \equiv \mathbf{T}$$

$$p \wedge \mathbf{F} \equiv \mathbf{F}$$

- De Morgan's laws:

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

- Absorption laws:

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

- Negations of $\mathbf{T}$ and $\mathbf{F}$:

$$\neg \mathbf{T} \equiv \mathbf{F}$$

$$\neg \mathbf{F} \equiv \mathbf{T}$$

Ex. 2.2.12 (Modified). Apply Ex. 2.2.13a (below) to establish the logical equivalence

$$p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r).$$

Next, use it rewrite the following statement. (Assume that x represents a fixed real number.)

If $x > 2$ or $x < -2$, then $x^2 > 4$.

p

q

r

(b) If $x > 2$ then $x^2 > 4$, and if $x < -2$ then $x^2 > 4$.



Ex. 2.2.13a. Use a truth table to verify the following logical equivalence. Include a few words of explanation with your answer.

$$p \rightarrow q \equiv \neg p \vee q.$$

SOLUTION .

p	q	$p \rightarrow q$	$\neg p \vee q$
1	1	1	1
1	0	0	0
0	1	1	1
0	0	1	1


same truth values

The table shows that $p \rightarrow q$ and $\neg p \vee q$ have the same truth values. Therefore, they are logically equivalent. □

Not In Syllabus – Propositional Logic

1. $p \rightarrow q$

Premise

2. $p \vee \neg p$

Law of Excluded Middle

3. p

Assumption

4. q

Modus Ponens 1,3

5. $\neg p \vee q$

Generalization 4

6. $\neg p$

Assumption

7. $\neg p \vee q$

Generalization 6

8. $\neg p \vee q$

Disjunction Elimination 2, 3-5, 6-7

Question E1. Are the following statement forms tautologies, contradictions, or neither?

- a. $p \wedge (p \vee \neg p)$
- b. $\neg(p \vee \neg p) \vee p$
- c. $\neg(p \wedge q) \vee p$

SOLUTION . a. $p \wedge (p \vee \neg p) \equiv p \wedge \mathbf{T} \equiv p$. (We used negation law and identity law). So neither tautology nor contradiction.

b. $\neg(p \vee \neg p) \vee p \equiv \neg\mathbf{T} \vee p \equiv \mathbf{F} \vee p \equiv p$. (We used negation law and identity law). So neither tautology nor contradiction.

c.

$$\begin{aligned}\neg(p \wedge q) \vee p &\equiv (\neg p \vee \neg q) \vee p && \text{(De Morgan's Law)} \\ &\equiv (\neg q \vee \neg p) \vee p && \text{(Commutative Law)} \\ &\equiv \neg q \vee (\neg p \vee p) && \text{(Associative Law)} \\ &\equiv \neg q \vee \mathbf{T} && \text{(Negation Law)} \\ &\equiv \mathbf{T} && \text{(Universal Bound Law)}\end{aligned}$$

It is a tautology.

